Geometric Aesthetics: A Framework for Beauty as Compressibility in Perceptual Eigenspaces

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***Abstract*—We propose a mathematical framework in which aesthetic experience—the perception of beauty, harmony, and elegance—arises from the compressibility of stimuli relative to a perceiver's learned perceptual eigenstructure. We formalize perceptual spaces as finite-dimensional inner product spaces equipped with a learned covariance, define a scoring functional that measures description efficiency relative to this structure, and derive three main results under simplifying assumptions: (1) the scoring functional is maximized at a characterizable boundary between order and randomness; (2) stimuli with group symmetry achieve exact compression as a special case; (3) a temporal extension yields a compression-progress functional whose peaks correspond to moments of "insight." We show how Birkhoff's aesthetic measure, Berlyne's inverted-U, and reward prediction error theory can be interpreted within this framework, and we derive heuristic predictions testable against existing datasets. The framework is proposed as a formalization of intuitions, not as a closed theory; we are explicit about where the mathematics is rigorous and where it is conjectural.**

***Index Terms*—Aesthetics, information theory, perception, symmetry, compression, eigenspectrum, beauty.**

I. Introduction

The question of why certain configurations are perceived as beautiful has occupied philosophers since antiquity and scientists since Fechner's experimental aesthetics in the 1870s [9]. Despite sustained inquiry, no mathematical framework has achieved broad acceptance.

We propose that aesthetic experience has a geometric structure that can be partially formalized. The central idea is simple: each perceiver carries a *perceptual eigenstructure*—a learned decomposition of the stimulus space into directions of decreasing importance—and the aesthetic response to a stimulus depends on the relationship between the stimulus's structure and the perceiver's eigenstructure. A stimulus that is compressible in the perceiver's eigenbasis but not in a random basis carries *meaningful structure relative to that perceiver*. We identify this property with beauty.

a) Scope and honesty.: This paper is a *conceptual theory paper with formal scaffolding*, not a theorem-driven paper in the classical sense. We prove some results rigorously (Propositions 4, 11, **??**, and 8). Other results are heuristic observations or conjectures, labeled as such (the temporal theory in §VI, broken symmetry in §V, connections to neuroscience and cultural distance in §VIII). We believe the framework is productive even where it is not yet rigorous, but we do not claim more rigor than exists.

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II. The Perceptual Eigenspace

A. Setup

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a *stimulus space*—the set of all possible stimuli in a domain. A perceiver P has been exposed to a distribution μ over $\mathcal{X}$ (their lifetime experience in the domain).

Definition 1 (Perceptual covariance)**.** The *perceptual covariance* of perceiver P is the positive semi-definite matrix

$$\Sigma_P = \mathbb{E}_\mu\big[(x-\bar{x})(x-\bar{x})^\top\big], \quad \bar{x} = \mathbb{E}_\mu[x]. \tag{1}$$

Its eigendecomposition $\Sigma_P = U\Lambda U^\top$ with $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$ defines the perceiver's *perceptual eigenbasis*.

Definition 2 (Perceptual coordinates)**.** The representation of stimulus x in the perceiver's eigenbasis is $z = U^\top(x-\bar{x}) \in \mathbb{R}^d$. In these coordinates, Σ_P is diagonal with entries λ_i.

Definition 3 (Participation ratio)**.** For a nonnegative spectrum $s_1, \ldots, s_d$ with $\sum s_i > 0$, the *participation ratio* is

$$D_{\text{eff}}(s) = \frac{(\sum_{i=1}^d s_i)^2}{\sum_{i=1}^d s_i^2}, \tag{2}$$

satisfying $1 \leq D_{\text{eff}} \leq d$. Applied to the eigenvalues, $D_{\text{eff}}(\lambda)$ measures the effective dimensionality of the perceiver's experience. Applied to $\{z_i^2\}$, it measures the effective dimensionality of a stimulus.

Proposition 4 (Expertise can expand effective dimensionality)**.** *Let $\Sigma' = (1-\alpha)\Sigma + \alpha\Sigma_{\text{new}}$ for $\alpha \in (0,1)$. If the normalized eigenvalue spectrum of Σ' majorizes that of Σ less sharply—that is, if the update* flattens *the normalized spectrum in the sense that $\sum_{i=1}^k \lambda_i'/S' \leq \sum_{i=1}^k \lambda_i/S$ for all $k < d$ (where $S = \sum \lambda_i$, $S' = \sum \lambda_i'$, and eigenvalues are sorted descending)—then $D_{\text{eff}}(\Sigma') \geq D_{\text{eff}}(\Sigma)$, with equality iff the normalized spectra coincide.*

Proof. The participation ratio is $D_{\text{eff}} = 1/\sum q_i^2$ where $q_i = \lambda_i/S$ is the normalized spectrum. The map $q \mapsto \sum q_i^2$ is a Schur-convex function on the probability simplex. By the Schur–Ostrowski criterion, $\sum (q_i')^2 \leq \sum q_i^2$ whenever q' is majorized by q (i.e., the normalized spectrum of Σ' is more uniform than that of Σ). Therefore $D_{\text{eff}}(\Sigma') \geq D_{\text{eff}}(\Sigma)$. Equality holds iff $q = q'$. □

Remark 5. The majorization condition is not automatically satisfied by adding variance on low-eigenvalue directions; it depends on the specific update. The proposition says that expertise *can* expand effective dimensionality—and does so precisely when the learning flattens the normalized spectrum. This is a sufficient condition, not a claim that all learning increases D_{eff}.

This formalizes the intuition that expertise in a domain expands the perceiver's effective perceptual dimensionality: a wine expert discriminates along dimensions invisible to a novice.

III. The Aesthetic Score

A. Motivation and Design

We seek a score $\mathcal{A}(x; \Sigma)$ that captures: *a stimulus is aesthetically valued when it carries meaningful structure relative to the perceiver*. "Meaningful structure" means the stimulus is (i) not trivial (it has content across multiple dimensions), (ii) not random (it is compressible), and (iii) compressible specifically in the perceiver's eigenbasis (the structure is aligned with learned experience).

B. Definition

We work on the unit energy surface $\|z\|^2 = 1$ to eliminate scale dependence. All stimuli are normalized to unit energy in perceptual coordinates before scoring. We restrict to the *active subspace*: let $r = |\{i : \lambda_i > 0\}|$ be the rank of Σ_P, and work in the r-dimensional subspace spanned by eigenvectors with positive eigenvalues. This avoids division by zero in the compressibility term.

Definition 6 (Aesthetic score)**.** Let $z \in \mathbb{R}^r$ with $\|z\|^2 = 1$, where r is the rank of Σ_P and all coordinates correspond to eigenvectors with $\lambda_i > 0$. Define the energy distribution $p_i = z_i^2$ (so $p_i \geq 0$ and $\sum_{i=1}^r p_i = 1$). Let $q_i = \lambda_i / \sum_{j=1}^r \lambda_j$ be the normalized perceiver eigenvalues. The *aesthetic score* is

$$\mathcal{A}(p; q) = \underbrace{D_{\text{eff}}(p)}_{\text{complexity}} \cdot \underbrace{\exp\left(-\frac{1}{r}\sum_{i=1}^{r}\frac{p_i}{q_i}\right)}_{\text{compressibility}}, \tag{3}$$

which is well-defined since $q_i > 0$ for all $i \leq r$.

The first factor, $D_{\text{eff}}(p) = 1/\sum p_i^2$, measures how many dimensions the stimulus occupies (complexity). The second factor measures how well the stimulus's energy distribution aligns with the perceiver's eigenvalue distribution: when p_i concentrates on high-λ_i directions, the argument of the exponential is small and the compressibility factor is close to 1; when energy is spread onto low-λ_i directions, the factor decreases.

The exponential form ensures the compressibility factor is bounded in $[0, 1]$, eliminating the sign and scale issues of the linear penalty used in the preliminary version.

Remark 7. On the unit simplex $\{p : p_i \geq 0, \sum p_i = 1\}$, the score $\mathcal{A}(p; \lambda)$ is well-defined, continuous, and bounded. At $p = e_i$ (all energy on one dimension), $D_{\text{eff}} = 1$ and $\mathcal{A}$ is small. At $p = (1/d, \ldots, 1/d)$ (uniform), $D_{\text{eff}} = d$ but the compressibility factor is generally small for non-uniform λ. The maximum occurs at an interior point of the simplex.

IV. The Aesthetic Optimum

Proposition 8 (Existence of an interior maximizer)**.** *The aesthetic score $\mathcal{A}(p; \lambda)$ on the closed simplex $\Delta^{d-1} = \{p : p_i \geq 0, \sum p_i = 1\}$ attains its maximum. The maximizer does not lie at any vertex $p = e_i$ (provided $d \geq 2$ and at least two $\lambda_i > 0$).*

Proof. $\mathcal{A}$ is continuous on the compact set Δ^{d-1}, so attains its maximum by the extreme value theorem. At any vertex $p = e_i$, $D_{\text{eff}}(e_i) = 1$, so $\mathcal{A}(e_i; \lambda) = 1 \cdot \exp(-1/(d \cdot \lambda_i/S))$ where $S = \sum \lambda_j$. By moving a small amount of mass δ from coordinate i to a coordinate j with $\lambda_j > 0$, D_{eff} increases to first order (since $\partial D_{\text{eff}}/\partial p_j|_{p=e_i}$ is unbounded from above as we leave the vertex), while the exponential factor changes continuously. Therefore e_i is not a maximizer. $\square$

Proposition 9 (Inverted-U on the power-law family)**.** *On the one-parameter family $p_i(\alpha) = c_\alpha \cdot i^{-\alpha}$ (where c_α normalizes to unit sum), the score $\mathcal{A}(\alpha) = \mathcal{A}(p(\alpha); \lambda)$ satisfies:*

(a) *$\mathcal{A}(0) > 0$ and $\mathcal{A}(\alpha) \to 0$ as $\alpha \to \infty$.*
(b) *There exists $\alpha^* > 0$ achieving the maximum.*
(c) *As $\beta \to 0$ (flat perceiver spectrum), $\alpha^* \to 0$.*

That is, the score is an inverted-U function of concentration (and hence of stimulus complexity).

Proof. (a) At $\alpha = 0$, $p_i = 1/d$ and $D_{\text{eff}} = d$, so $\mathcal{A}(0) = d \cdot \exp(-S^{-1} \sum \lambda_i^{-1}/d^2) > 0$. As $\alpha \to \infty$, all mass concentrates on $i = 1$, so $D_{\text{eff}} \to 1$ and $\mathcal{A} \to \exp(-1/(d \cdot \lambda_1/S))$, which is less than $\mathcal{A}(0)$ for $d \geq 2$.

(b) By continuity of $\mathcal{A}(\alpha)$ on $[0, \infty)$ and the boundary behavior in (a), the maximum is attained at some $\alpha^* \in (0, \infty)$.

(c) When $\beta = 0$ (all λ_i equal), the compressibility factor is constant for all p, so $\mathcal{A}$ is proportional to $D_{\text{eff}}(p)$, which is maximized at $\alpha = 0$. $\square$

This formalizes Berlyne's [3] central finding: the optimal stimulus complexity is intermediate.

Remark 10 (What we do not prove). We do not prove that $\alpha^* < \beta$ in general. Numerical experiments suggest this holds for moderate β (the aesthetically optimal stimulus decays more slowly than the perceiver's spectrum), but for sharply peaked spectra ($\beta \gtrsim 2$ at moderate d) the maximizer can occur at $\alpha > \beta$. We also do not prove uniqueness of α^* or characterize the global optimum over the full simplex. These remain open problems.

V. Symmetry as Compression

The classical theory of beauty as symmetry is a special case of compression.

Definition 11 (Fixed-point subspace)**.** Let G be a finite group acting on $\mathbb{R}^d$ via linear transformations $\{T_g\}_{g \in G}$. The *fixed-point subspace* is $\mathrm{Fix}(G) = \{v \in \mathbb{R}^d : T_g v = v \ \forall g \in G\}$.

Proposition 12 (Symmetry implies compression)**.** *If $x \in \mathrm{Fix}(G)$ for some representation of G on $\mathbb{R}^d$, then x is determined by $\dim(\mathrm{Fix}(G))$ coordinates. The compression ratio relative to the ambient space is $d/\dim(\mathrm{Fix}(G))$.*

Proof. $\mathrm{Fix}(G)$ is a linear subspace of $\mathbb{R}^d$. Any element $x \in \mathrm{Fix}(G)$ can be represented in a basis for this subspace using $\dim(\mathrm{Fix}(G))$ coordinates. The compression ratio follows by definition. □

Remark 13. The dimension $\dim(\mathrm{Fix}(G))$ depends on the specific representation, not just on $|G|$. For the regular representation, $\dim(\mathrm{Fix}(G)) = 1$ (maximal compression). For a representation with many trivial summands, $\dim(\mathrm{Fix}(G))$ can be much larger. Specific examples:

Example 14 (Bilateral symmetry). For $G = \mathbb{Z}/2\mathbb{Z}$ acting on $\mathbb{R}^d$ by $(z_1, z_2, \ldots, z_{d/2}, z_{d/2+1}, \ldots, z_d) \mapsto (z_{d/2+1}, \ldots, z_d, z_1, \ldots, z_{d/2})$ (swapping left and right halves), $\mathrm{Fix}(G)$ consists of vectors with identical halves: $\dim(\mathrm{Fix}(G)) = d/2$, giving $2:1$ compression.

Example 15 (Rotational symmetry of order n). For $G = \mathbb{Z}/n\mathbb{Z}$ acting by cyclic permutation on $\mathbb{R}^n$, $\mathrm{Fix}(G)$ consists of constant vectors: $\dim(\mathrm{Fix}(G)) = 1$, giving $n:1$ compression.

a) Broken symmetry (heuristic).: The intuition that perfect symmetry is boring—and that small asymmetries can enhance aesthetic value—is well-supported empirically (slightly asymmetric faces are preferred over perfectly symmetric ones). Within our framework, this corresponds to the observation that a small perturbation $\epsilon \perp \mathrm{Fix}(G)$ added to a symmetric stimulus $x_s \in \mathrm{Fix}(G)$ can increase D_{eff} (by activating new dimensions) faster than it decreases the compressibility factor (by spreading energy onto less-aligned directions).

However, whether $\mathcal{A}(x_s + \epsilon) > \mathcal{A}(x_s)$ depends on the specific directions in which ϵ lies. If the perturbation projects onto directions with very small λ_i (directions the perceiver considers unimportant), the compressibility penalty can dominate. A general proposition would require an explicit condition on the alignment of ϵ with the perceiver's eigenspectrum. We leave this as an open problem and note that the phenomenon is robustly observed in the empirical literature on symmetry preference [5].

VI. Compression Progress: A Temporal Extension

Many aesthetic experiences are temporal: music, narrative, mathematical proof. We propose a temporal extension, noting that this section is more speculative than the preceding ones.

A. Setup

Let $x_1, x_2, \ldots, x_T \in \mathbb{R}^d$ be a sequence of stimuli. We represent the perceiver's cumulative experience at time t by the sufficient statistic

$$\bar{z}_t = \frac{1}{t}\sum_{s=1}^{t} z_s, \qquad S_t = \frac{1}{t}\sum_{s=1}^{t} z_s z_s^\top, \tag{4}$$

where $z_s = U^\top(x_s - \bar{x})$ are perceptual coordinates. The sample covariance is $\hat{\Sigma}_t = S_t - \bar{z}_t \bar{z}_t^\top$.

Definition 16 (Compression progress)**.** Let $p_s^{(t)}$ denote the normalized energy distribution of z_s relative to the eigenbasis of $\hat{\Sigma}_t$: that is, $p_{s,i}^{(t)} = (u_i^{(t)\top} z_s)^2 / \|z_s\|^2$ where $\{u_i^{(t)}\}$ are the eigenvectors of $\hat{\Sigma}_t$, and let $q^{(t)}$ be the normalized eigenvalues of $\hat{\Sigma}_t$. The *compression progress* at time t is

$$\dot{\mathcal{A}}_t = \sum_{s=1}^{t} \mathcal{A}(p_s^{(t)}; q^{(t)}) - \sum_{s=1}^{t} \mathcal{A}(p_s^{(t-1)}; q^{(t-1)}). \tag{5}$$

This measures how much better the perceiver can describe all prior stimuli using the updated model versus the previous model.

In words: compression progress measures how much better the perceiver can describe *all data seen so far* using their updated model. A large positive value at time t means x_t caused a model update that retroactively made everything more compressible—a moment of insight.

Conjecture 17 (Compression progress under rank-one updates)**.** *If $\hat{\Sigma}_t$ differs from $\hat{\Sigma}_{t-1}$ by an approximate rank-one update (the new stimulus is the dominant source of change), then $\dot{\mathcal{A}}_t$ is approximately proportional to the squared projection of z_t onto the eigenvector of $\hat{\Sigma}_{t-1}$ that undergoes the largest relative eigenvalue increase.*

In particular, compression progress is large when the new stimulus (i) projects strongly onto a previously low-variance direction, and (ii) reveals a systematic pattern (not just noise) that makes prior data more compressible along that direction.

We do not prove this conjecture in full generality. The difficulty is that the aesthetic score $\mathcal{A}$ involves the participation ratio, which is not differentiable at certain boundaries and does not decompose cleanly under rank-one updates. A rigorous treatment would require smoothing or restricting to the power-law family. We include the conjecture because it generates testable predictions and connects to the neuroscience literature.

Example 18 (The punchline of a joke). A joke's setup builds a model (high-λ directions). The punchline reveals an alternative interpretation: a previously zero-variance direction suddenly explains the setup data. The compression progress is large because the entire setup sequence becomes more compressible under the new model.

Example 19 (Modulation in music). A modulation from the home key to a distant key projects onto a previously low-variance direction in harmonic space. The return to the home key causes another spike: the listener's model updates to accommodate the overall tonal plan.

VII. Connections to Classical Theories

We now show how the framework *relates to* (not "subsumes") three classical theories.

a) Birkhoff's aesthetic measure.: Birkhoff [2] defined $M = O/C$ (order over complexity). In our framework, order corresponds to compression via symmetry (Proposition 11) and complexity to D_{eff}. The ratio O/C is a monotone function of the compressibility-per-dimension, which is one component of our score. Our framework replaces the division (which penalizes complexity) with a product (which rewards complexity up to a point), addressing Birkhoff's known failure to account for the appeal of complex stimuli.

b) Berlyne's inverted-U.: Corollary 8 derives the inverted-U as a consequence of the competing drives toward complexity and compressibility. The novel prediction is that the peak shifts with perceiver eigenvalue distribution, which is measurable through preference experiments stratified by expertise.

c) Reward prediction error.: The neuroscience literature reports that aesthetic pleasure correlates with dopaminergic prediction error [8]. We conjecture (but do not prove) that compression progress (Definition 16) is a computational-level description of the same signal: the brain's reward system fires when the internal model improves—when something that was complex becomes compressible. This interpretation follows Schmidhuber [4], who first proposed compression progress as a theory of beauty. Our contribution is to ground it in a specific eigenspace framework rather than abstract Kolmogorov complexity.

VIII. Conjectural Extensions

We briefly note two extensions that are speculative but suggestive.

a) Cultural distance.: The space of positive definite matrices $\mathcal{S}^d_{++}$ carries the Fisher information metric, under which the geodesic distance between Σ_A and Σ_B is $d_F = \left\|\log(\Sigma_A^{-1/2}\Sigma_B\Sigma_A^{-1/2})\right\|_F$. We conjecture that aesthetic disagreement between cultural groups A and B on a stimulus x is bounded by a Lipschitz multiple of $d_F(\Sigma_A, \Sigma_B)$. This would formalize the intuition that cultures with similar artistic traditions have similar aesthetic preferences. A proof would require careful handling of the coordinate-dependent nature of the score.

b) Cross-domain universality.: If the same perceiver has eigenstructures in different domains (visual, musical, gustatory), the framework predicts that aesthetic principles are *structurally identical* across domains but the *eigenspaces are different*. Symmetry is beautiful in faces and in music for the same reason (compression), but a visual art expert's expanded eigenbasis does not transfer to music appreciation. This is testable.

IX. Predictions

The framework generates heuristic predictions, all testable against existing or obtainable data:

1) **Expertise shift.** Experts prefer stimuli with higher effective dimensionality than novices (Corollary 8). Testable by comparing preferred complexity in trained vs untrained listeners.
2) **Broken symmetry preference.** Stimuli with small controlled asymmetry should be preferred over both perfect symmetry and strong asymmetry, provided the perturbation aligns with perceiver-important directions. Testable against face preference data with parametric asymmetry manipulation.
3) **Compression progress predicts chills.** Musical "frisson" should correlate with peaks of $\dot{\mathcal{A}}_t$ (Conjecture 17). Testable by computing the score from MIDI and comparing against physiological chill recordings.
4) **Mathematical elegance.** Among proofs of the same theorem, mathematicians should rate shorter proofs (lower description length) as more elegant. Approximately testable using proof length in a fixed formal system.
5) **No cross-domain transfer.** Training in visual art should not shift preferred complexity in music, because the eigenspaces are domain-specific. Testable via a pre/post design with domain-specific preference measurement.

X. Discussion

a) What is rigorous.: Propositions 4 (under the majorization condition), 11, **??**, and 8 are proved. The existence of an interior maximizer and the inverted-U shape on the power-law family are established; the location and uniqueness of the maximizer are not.

b) What is heuristic.: The aesthetic score (Definition 5) is a motivated construction, not derived from first principles. The exponential compressibility factor is chosen for boundedness and mathematical convenience; other monotone, bounded functions of the alignment penalty would yield qualitatively similar behavior. The broken symmetry phenomenon (§V) is a heuristic observation supported by empirical evidence but not proved in full generality.

c) What is conjectural.: The temporal theory (Conjecture 17), the cultural distance bound, the connection to reward prediction error, and the cross-domain universality claim are interpretive connections, not proved results. We include them because they generate testable predictions and point toward a research program.

d) Linearity.: We model perceptual spaces as linear. Real perceptual spaces are likely nonlinear manifolds. Extending to kernel PCA or autoencoders would capture nonlinear structure at the cost of analytical tractability. The qualitative predictions (inverted-U, expertise shift, broken symmetry) should survive nonlinearization, but the closed-form results will not.

e) The PCA-Matryoshka connection.: In PCA-Matryoshka [1], we showed that eigenvalues serve as importance weights for embedding compression: truncating in the PCA basis preserves meaning while truncating in a random basis destroys it. The aesthetic framework developed here suggests this may reflect something deeper about perception. Beauty may be the subjective experience of encountering meaningful structure—structure that compresses in the basis evolution and experience gave us to understand the world.

This connection is metaphorical, not proved. But we find it suggestive enough to state.

XI. Conclusion

We have proposed a mathematical framework for aesthetic experience based on compressibility in perceptual eigenspaces. The framework yields four proved results (expertise expands effective dimensionality under a majorization condition; symmetry compresses description length; an interior maximizer exists; the score is inverted-U on the power-law family), one heuristic observation (broken symmetry), one conjecture (compression progress for temporal aesthetics), and five testable predictions.

The framework interprets—but does not subsume—Birkhoff's aesthetic measure, Berlyne's inverted-U, and reward prediction error theory. The main value, we believe, is not in the specific functional form chosen, but in the conceptual identification: *the perceiver's learned eigenstructure determines what counts as meaningful structure, and beauty is the perception of structure that is meaningful in this sense*.

The framework is incomplete. A complete theory of aesthetics would need to account for temporal dynamics beyond our conjecture, social and cultural mechanisms beyond our Fisher metric speculation, emotional valence beyond our compression-based proxy, and the irreducible role of personal history and association. We offer this framework not as a final word but as a mathematical starting point that is precise enough to be wrong in testable ways.

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